

FACULTY OF SCIENCE, AGRICULTURE & ENGINEERING**DEPARTMENT OF ENGINEERING**

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DATE SUBMITTED	19 / 04 / 2026		

GAs STUDENT SUBMISSION COVER PAGE**Signed:**

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Appendix A

1. AI Tool Use Disclosure

For each AI tool that you have used, provide a brief, high-level summary:

1.	Tool Name and Version:	ChatGPT
	Purpose:	To help explain a bit of what the topic is about.
	Scope of Use	To explain the topic.
2.	Tool Name and Version:	DeepSeek AI
	Purpose:	To help learn how ladder logic works.
	Scope of Use	To explain ladder logic.

2. AI Tool Use Detailed Disclosure by Category

For every instance where you used an AI tool, you must provide a detailed explanation.

A. AI as a Writing Tool (*e.g., drafting, summarising, paraphrasing, editing, proofreading, image generation*)

Requirement	Explanation
Specific use	<i>Abstract pg. 4</i>
Input/Prompt	Help me summarise this abstract to only include the most important information.
Validation process	I used AI to help summarise my abstract and make it more direct.

Requirement	Explanation
Specific use	<i>Introduction pg.6</i>
Input/Prompt	Here is what I wrote, please help me edit for correct grammar.
Validation process	I used AI to help me improve my grammar.

Requirement	Explanation
Specific use	<i>Field Components pg. 7</i>
Input/Prompt	Reword this paragraph so it's not the same as the references.
Validation process	To help me reword the work so that I don't plagiarise any work.

AI as an Idea Generation Tool (*e.g., brainstorming, outlining, topic selection, structural design*)

Requirement	Explanation
Specific use	<i>Calculations pg. 11</i>
Input/Prompt	Give me an idea of what to include in calculations section of the report.
Validation process	To help understand what other calculations I should be doing.

Requirement	Explanation
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Specific use	<i>Control objectives pg. 10</i>
Input/Prompt	Help me understand what the control objectives for this topic are and how to map them out.
Validation process	To help understand the control objectives.

Requirement	Explanation
Specific use	<i>I/O Mapping pg.7</i>
Input/Prompt	Based on my idea of this topic, generate a simple flow chart of how I explained the system should work.
Validation process	To help develop a cleaner and improved sequence flow chart by myself.

B. AI as a Code Generation or Development Tool (e.g., generating code snippets, debugging, explaining code, writing test cases, any other technical/implementation aspects in your design project)

Requirement	Explanation
Specific use	<i>Ladder logic pg. 13</i>
Input/Prompt	Please explain in detail how to use timers in ladder logic and how to use Reset in ladder logic.
Validation process	To help gain a deeper understanding on how to use the component for a specific function in the logic diagram.

Requirement	Explanation
Specific use	<i>Ladder logic pg. 13</i>
Input/Prompt	Please explain in detail how to build my ladder logic in the picture I uploaded into the software Codesys.
Validation process	To help understand how to implement my logic diagram using Codesys functions.

Abstract:

This report focuses on the design of a PLC-based control system for an automated fertilizer batch blending process. The system is used to blend Nitrogen (N), Phosphorus (P), and Potassium (K) using timer-controlled feeders, followed by mixing and discharge through a conveyor. Different inputs such as load cells, temperature and moisture sensors and conveyor speed are used to monitor the process. The PLC controls the ingredient feeders, mixer motor, output conveyor, and alarm system. The ladder logic was simulated to verify the operation, and the results show that the system improves accuracy, safety, and efficiency compared to manual blending.

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1. Introduction:

Omnia Holdings Limited is a group that researches, manufactures and supplies chemicals and specialized services for agriculture, mining and chemical application industries [1]. One of their key operations is the production of fertilizers at the plant in Sasolburg, in the Free State province of South Africa. Fertilizers are essential products used in agriculture to improve soil nutrients and increase crop size [2].

The fertilizer blending process works by mixing three key ingredients which is Nitrogen (N), Phosphorus (P), and Potassium (K) in specific amounts to make different types of compound fertilizer. Traditionally, this was done by hand, which caused a lot of problems including human error, uneven chemical ratios, slow production, and safety risks for the workers dealing with these bulk chemicals. To solve these problems, modern plants now use Programmable Logic Controllers (PLCs) to automate the blending process.

A PLC is a tough industrial computer built to control manufacturing equipment. In this system, the PLC takes in signals from sensors and devices in the field like load cells, temperature and moisture sensors, and emergency stop switches and uses that information to control feeders, motors, and conveyors.

This report covers the design of a PLC control system for the Omnia Holdings batch blending plant in Sasolburg. The system uses nine inputs (three load cells for N, P, and K, a mixer motor overload, temperature and moisture sensors, conveyor speed, an emergency stop, and a batch start switch) and six outputs to control the feeders, mixer motor, conveyor, and alarm to making sure every fertilizer batch is produced consistently and safely.

2. Process Description:

The fertilizer blending process consists of operational phases:

- The batch start button is pressed to start the process. An emergency stop button is also placed in case of failure.
- Once the batch start button is activated, it starts with the T1 timer which times the Nitrogen load cell being filled, after the time passes, it will trigger the Nitrogen feeder to stop.
- When the Nitrogen feeder signal becomes TRUE, it triggers the Potassium timer T2 to begin filling the Potassium load cell. After the time passes, it will trigger the Potassium feeder to stop and for the Phosphorus timer to begin.
- After the Phosphorus timer T3 passes for filling the load cell, it will trigger the Phosphorus feeder to stop.
- Ensuring that temperature and moisture levels are adequate, the mixer motor will begin the mixing process.
- The conveyor speed can be adjusted to trigger the conveyor.
- An emergency alarm can be triggered for any faults with the loading cells and if the mixer motor overloads that stays on until the alarm is switched off by a worker. When a fault triggers the alarm, it automatically switches the mixer motor and conveyor off.

3. Field Components:

3.1.) Temperature Sensor:

Temperature inside the mixer is important because fertilizer materials can react chemically when overheated. Increased temperature can cause moisture evaporation, chemical degradation, or clumping of materials. Temperature sensors such as thermocouples or RTDs operate based on changes in voltage with temperature [3]. The PLC monitors this value by the sensor to prevent overheating and ensure stable blending conditions.

3.2.) Load Cells (N, P, K Measurement):

Strain gauges are used in load cells. Load cells measure the weight of each ingredient using strain measurement. When the chemical is added, the load cell slightly deforms. Using strain gauge provides reliable and accurate measurement [3][6].

3.3.) Moisture Sensor:

Moisture level affects the physical properties of fertilizer. High moisture causes clumping, poor mixing, and blockages in feeders, while low moisture may cause dust formation. Moisture sensors use capacitive measurement due to its high reliability and accuracy in detecting small changes in moisture. The PLC uses this information to stop or alarm when moisture is not within acceptable limits [7][8].

4. Sequence flow chart:

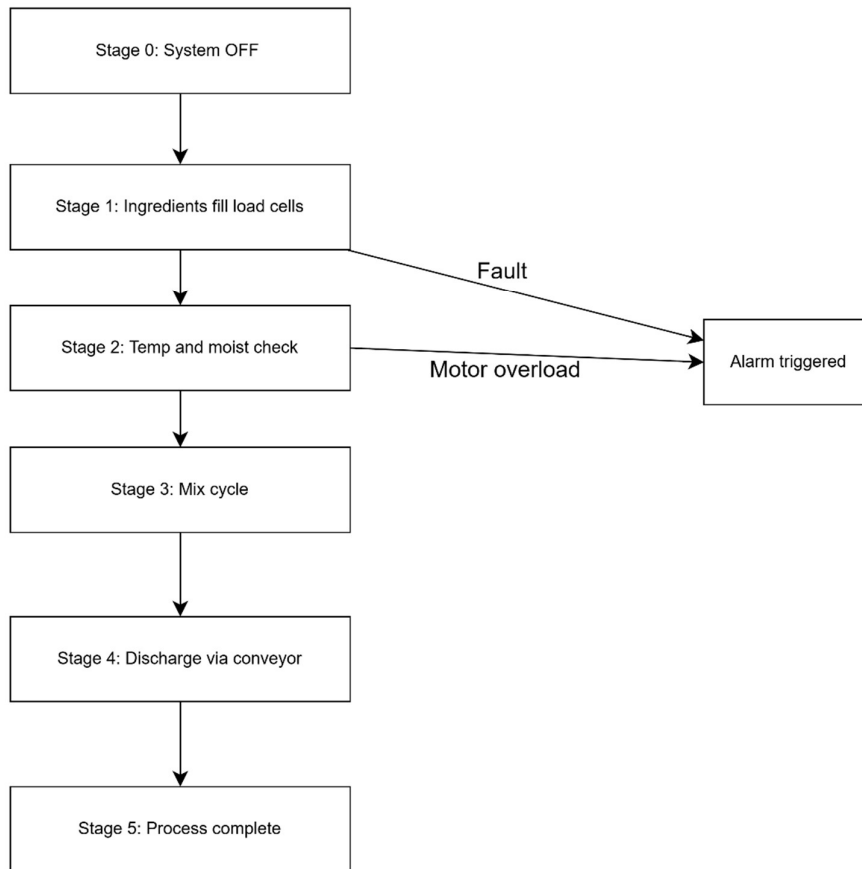


Figure 1: Sequence flow diagram of fertilizer batch blending system

The sequence flow diagram shows how the PLC moves from one stage to the next in the fertilizer blending process. Starting in the OFF state, the system waits for the operator to press the batch start button. Once activated, all three ingredient feeders begin filling their load cells with Nitrogen, Phosphorus, and Potassium one after the other. When all three ingredients have been filled, the PLC checks that the temperature and moisture levels inside the mixer are within the safe range before allowing the process to continue. I decided to place these checks after ingredient loading rather than before, because the act of adding the chemicals can itself affect the internal conditions of the mixer, so checking afterwards gives a more accurate reading. The mixer motor switches on to begin the mixing cycle. If any fault is detected at any point such as a motor overload or an out-of-range sensor then the alarm is triggered and the process stops until the operator fixes it. Once everything is sorted, the conveyor turns on to distribute the finished product, and the system returns to off stage and ready for the next batch.

5. PLC Inputs and Outputs:

There are a total of 9 inputs and 6 outputs where the inputs are three ingredient load cells (N, P, K), mixer motor overload, temperature sensor, moisture sensor, conveyor speed, emergency stop, and batch start switch. Output controls consist of three ingredient feeders, mixer motor, output conveyor, and alarm.

Table 5-1: Table of inputs with addresses

Inputs:	Description:	Address:
Batch Start	Button to start the fertilizer mixing process.	X0
Emergency Stop	Button to stop the whole system in case of faults and errors.	X1
Nitrogen Load cell	Switch that closes when the loading timer is completed.	X2
Phosphorus Load cell	Switch that closes when the loading timer is completed.	X4
Potassium Load cell	Switch that closes when the loading timer is completed.	X6
Temperature sensor	Switch closes when the measured temperature is within the acceptable range.	X8
Moisture sensor	Switch closes when the measured moisture is adequate for mixing.	X9
Conveyor speed	Switch closes when conveyor reaches acceptable speed.	X11
Mixer Motor overload	Switch opens when the mixer motor overloads.	X10
Memory for batch start	Memory is used to latch to keep the batch start switch on.	M0
Fault button	Fault button to switch off the Nitrogen, Phosphorus and Potassium feeders.	X3, X5, X7
Alarm switch	Switch to turn the alarm off after the alarm has been triggered.	X13

Table 5-2: Table of outputs with addresses

Outputs:	Description:	Address:
Nitrogen Feeder	Feeds Nitrogen into load cell.	Y0
Phosphorus Feeder	Feeds Phosphorus into load cell.	Y1
Potassium Feeder	Feeds Potassium into load cell.	Y2
Memory for batch start	Memory is used to latch to keep the batch start switch on.	M0
Memory for Mixer motor	Memory is used to latch to keep the mixer motor on.	M2
Mixer motor	Mixer used to mix the chemicals.	Y3
Conveyor	Conveyor used to distribute the chemicals after mixing.	Y4
Alarm	Buzzer to notify when there is a fault and error.	Y5

6. Control Objectives:

The fertilizer blending plant uses a PLC to automatically control the mixing of three ingredients into a fertilizer blend. Instead of using compare functions to measure exact weights, the system uses timers to run each feeder for a set amount of time. This means each feeder runs for a fixed duration, and when the timer is done, the feeder stops. The PLC then checks temperature and moisture conditions before starting the mixer and discharges the final product.

6.1. Safety and Master Enable:

The PLC checks three conditions before allowing the batch to begin:

- The emergency stop must not be pressed.
- The batch start button must be pressed by the operator.
- The mixer motor overload button must show no fault.

If all three conditions are met, the PLC operates as designed. If any condition fails, the system will not start and the alarm will activate.

6.2. Ingredient Loading Using Timers:

Once the system is on, the three ingredient feeders are started. Instead of monitoring weight and stopping when a target is reached, each feeder is controlled by a timer. The timer is pre-

set to the time needed to dispense the correct amount of each ingredient into the mixer since this approach assumes the filling rate of each chemical is constant.

- The Nitrogen feeder runs for a pre-set time, then stops automatically when its timer is done.
- The Phosphorus feeder runs for its pre-set time and stops when its timer is done.
- The Potassium feeder runs for its pre-set time and stops when its timer is done.

All three feeders run one after the other. The system only moves to the next stage once all three timers have completed, confirming that all ingredients have been dispensed into the mixer.

6.3. Temperature and Moisture Monitoring:

After the ingredients have been loaded, the PLC checks that the conditions inside the mixer are acceptable before starting the mixing process.

- A temperature switch monitors the mixer temperature. If the temperature is within the safe operating range, the temperature button is activated.
- A moisture switch monitors the moisture level inside the mixer. If the moisture is within the acceptable range, the moisture button is activated.
- If either condition is out of range, the alarm is activated until the alarm is switched off.

6.4. Mixer Motor Control:

Once all ingredients are loaded and conditions are confirmed as safe, the PLC starts the mixer motor. The mixer motor will only start if the following are all true at the same time:

- All three feeder timers have completed.
- Temperature is within range.
- Moisture is within range.

6.5. Conveyor Control:

Once the mixer is running, the PLC checks the output conveyor speed before allowing the blended product to be discharged. The conveyor speed switch confirms the belt is running at the required speed.

- If the belt is at the correct speed, the output conveyor is activated and the product is discharged.
- If the belt is not yet up to speed, the system waits until the speed switch confirms the belt is running correctly.

6.6. Fault Detection and Alarm:

The alarm is activated if any one of the following faults occur at any time:

- An error in the load cells.
- The mixer motor overloads.
- The temperature rises above the acceptable range.

7. Calculations:

1. Timer Preset Calculations:

Since Delta PLC timers use a 100 ms default by default [3], therefore the preset value K is:

$$K = \text{Required time (seconds)} \times 10$$

Table 7-1: Table of preset value calculations:

Timer:	Component purpose:	Time (sec):	Calculation:	Preset value (K):
T1	Nitrogen feeder fills load cells.	5 seconds	$5 \times 10 = 50$	K50
T2	Phosphorus feeder fills load cells.	3 seconds	$3 \times 10 = 30$	K30
T3	Potassium feeder fills load cells.	2 seconds	$2 \times 10 = 20$	K20

The Delta PLC timer uses a default time base of 100 ms [3]. Therefore, to achieve the time for the Nitrogen feeder to fill the load cells for 5 seconds, the preset value is calculated as $K = 5 \times 10 = K50$. The Phosphorus feeder to fill the load cell will be for 3 seconds, therefore the preset value is calculated as $K = 3 \times 10 = K30$. The Potassium feeder will fill the load cell for 2 seconds, therefore the preset value is calculated as $K = 2 \times 10 = K20$.

2. Signal Scaling Formula:

The sensors used in this system such as the temperature sensor, moisture sensor, load cells, and conveyor speed, output a standard 4–20 mA signal. This is the industry standard for process instrumentation in industrial automation applications, as it provides noise immunity over long cable runs and allows for wire-break detection since a live circuit will always carry a minimum of 4 mA [4][5].

The PLC cannot read milliamps directly and therefore, it converts it to a raw number between 0 and 4000.

3. Threshold Values:

The fertilizer batch blending system operates within defined limits to ensure safe and consistent production. Each sensor in the system monitors a specific physical parameter and the PLC continuously compares the measured value with the threshold. If any parameter exceeds its safe operating limit, the PLC will activate the alarm and stop the affected part of the process to prevent equipment damage, product quality failure, or harm to workers. The threshold values used in this system are summarised in the table below.

Table 7-2: Table of thresholds for sensors

Parameter:	Safe Range:	Alarm triggers when:
Temperature	15°C and 55°C	Below 15°C and above 55°C.
Moisture	0% – 40%	Above 40%

4. Feeder Dispensing Rate:

Since the three ingredient feeders in the batch blending system are controlled by timers rather than continuous weight monitoring, each feeder runs for a certain time, and when the time elapses, the feeder is automatically stopped by the PLC. The dispensing rate of each feeder is assumed to be constant throughout the filling process.

Using the relationship between mass, rate, and time, the required timer duration for each feeder can be calculated as follows:

$$\text{Time} = \text{Mass} \div \text{Rate}$$

For demonstration purposes, the Nitrogen feeder is assumed to dispense at a rate of 40 kg/s with a target load of 200 kg, the Phosphorus feeder dispenses at 30 kg/s with a target load of 90 kg, and the Potassium feeder dispenses at 20 kg/s with a target load of 40 kg. The timer preset values are therefore calculated as:

- T1 (Nitrogen): Time = $200 \div 40 = 5$ seconds, therefore K50.
- T2 (Phosphorus): Time = $90 \div 30 = 3$ seconds, therefore K30.
- T3 (Potassium): Time = $40 \div 20 = 2$ seconds, therefore K20.

8. PLC Ladder Logic:

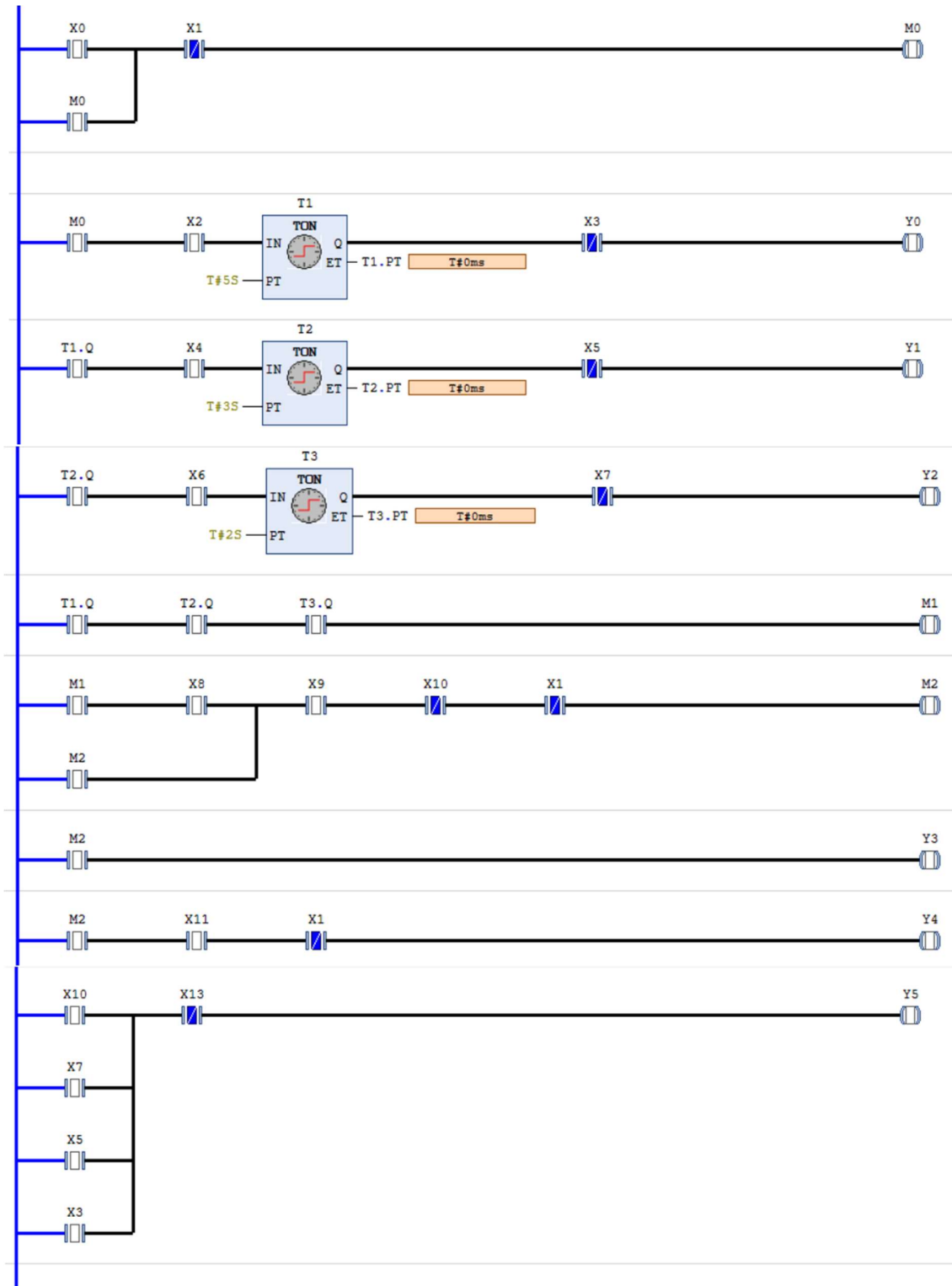


Figure 2: PLC Ladder Logic for fertilizer batch mixing process

9. Conclusion:

The PLC-based control system for the Omnia Holdings fertilizer blending plant in Sasolburg was successfully designed and tested. The system automates the blending of Nitrogen, Phosphorus, and Potassium using nine inputs and six outputs, and the ladder logic simulation was implemented successfully and responds correctly under both normal and fault conditions.

Safety was an important part of the design. The emergency stop, motor overload protection, and alarm system all work together to shut the process down if a fault or error is detected, and the alarm stays on until a worker manually turns it off, to ensure faults are not ignored. One limitation of the design is that the timer-based feeding assumes each feeder dispenses at a constant rate. But in reality, factors like the material properties and material clumping could affect how much is actually dispensed.

Despite this, the system does successfully meet all the control objectives set out in this report and is a great solution for the challenges faced in manual blending process.

10. References:

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